

A Novel Approach for Detecting a Stochastic Gravitational Waves Background

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Cosmic Microwave Background

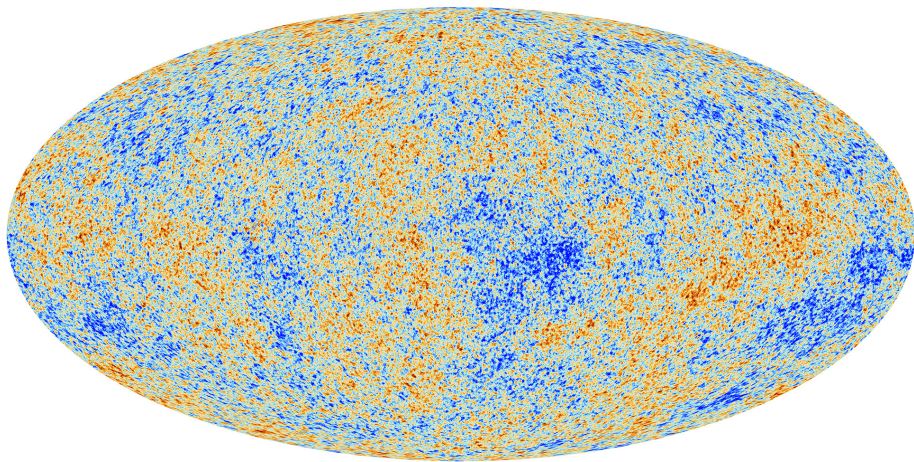


Figure: CMB map from Planck. Credit: ESA and the Planck Collaboration.

Gravitational Waves

Theory

Assumptions

- Gravity is a classical spin-2 field theory governed by the non-linear Einstein equations $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \frac{1}{M_{pl}^2}T_{\mu\nu}$.
- All gravitational fields are weak such that we can write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and work to first order in $h_{\mu\nu}$.

Properties

- The vacuum Einstein equations reduce to wave equations and permit two independent propagating wave modes. We will call these $+$ and $\times$.
- Gravitational Waves (GWs) propagate at the speed of light without dispersion.

Gravitational Waves

Observation

- GWs perturb position of objects.
- Take example + GW metric.

$$ds^2 = -dt^2 + dz^2 + [1 + h \cos(\omega t - kz)]dx^2 + [1 - h \cos(\omega t - kz)]dy^2$$

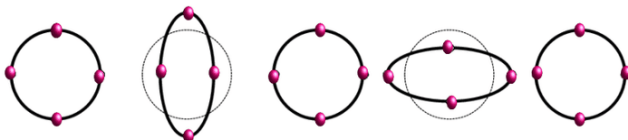


Figure: Plus polarisation GW metric perturbation. Credit: Imène Belahcene.

Gravitational Waves

Observation



Figure: LIGO Hanford. Credit: Caltech/MIT/LIGO Lab.

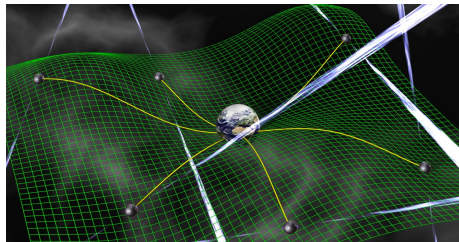


Figure: Pulsar Timing Array. Credit: David Champion.

Observables

- Need to define observables to predict and measure.
- Define scalar observable z as PTA redshift or blueshift.
- Consider astrometric distortion $l_{obs}(\hat{n}) = l_{true}(\hat{n} + \delta\hat{n})$.
- Also define distortion matrix $\psi_{ab} = \frac{d(\delta\hat{n}_a)}{d\hat{n}^b}$.
- Can SVT decompose to get new observables.
- Have two scalar observables κ, ω , two vector observables $\delta_{\pm}$ and two tensor observables $\pm\gamma$.
- Need to know response functions to accurately predict observables.

Correlation Functions

Start with general GW mode decomposition

$$h_{ij}(t, \mathbf{x}) = \sum_P \int df \int d^2 \hat{q} \, h_P(f, \hat{q}) \epsilon_{ij}^P(\hat{q}) e^{2\pi i f(t + \hat{q} \cdot \mathbf{x})}$$

and promote $h_P(f, \hat{q})$ to Gaussian random variable obeying

$$\langle h_P(f, \hat{q}) h_{P'}^*(f', \hat{q}') \rangle = \delta_{PP'} \delta(f - f') \delta^2(\hat{q} - \hat{q}') H(f) .$$

Can find correlator between any two observables

$$\langle A(\hat{n}_1, t) B^*(\hat{n}_2, t') \rangle = \Gamma^{AB}(\hat{n}_1, \hat{n}_2) K(t - t') .$$

Correlation Functions

Hellings-Downs Curve

- Most famous example of SGW correlator.
- $\langle zz \rangle \sim \Gamma^{zz}$ correlation function.

$$\Gamma^{zz}(\xi_{ab}) = \frac{1}{3} + \frac{1}{2}(1 - \cos \xi_{ab}) \left[-\frac{1}{6} + \log \left(\frac{1 - \cos \xi_{ab}}{2} \right) \right]$$

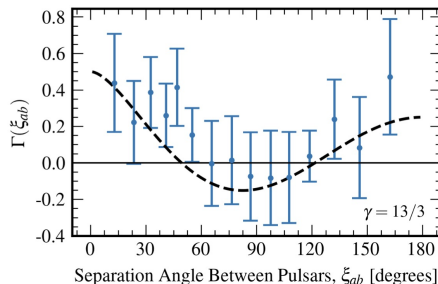


Figure: Hellings-Downs curve with data. Credit: NANOGrav collaboration.

Correlation Functions

Short-Distance Correlators (arXiv:2311.03474)

- Recent development by Mentasti & Contaldi to observe GWs inside solar system by astrometric measurements of asteroids.
- Can compute simple $\cos$ correlation functions for short-distance limit.

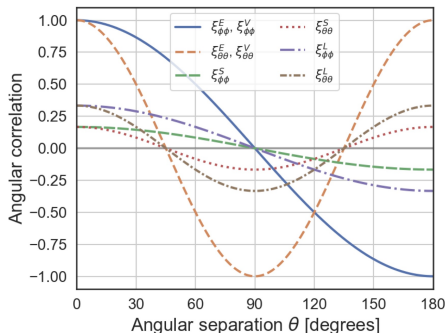


Figure: Short-distance correlators. Credit: Giorgio Mentasti.

Spin-Weighted Formalism

- Introduce functions with spin-weight s .
- Functions with spin-weight s transform as ${}_s f(\theta, \phi) \rightarrow e^{is\theta} {}_s f(\theta, \phi)$ under rotation around north pole.
- Can decompose by defining generalisation of spherical harmonics with spin-weight ${}_s Y_{lm}(\theta, \phi)$.
- Introduce spin raising $\bar{\partial}$ and lowering ∂ operators.
- Can expand scalar ($s = 0$), vector ($s = 1$) and tensor ($s = 2$) observables in ${}_s Y_{lm}(\theta, \phi)$.
- Lower spin-weight of function to $s = 0$, expand in spherical harmonics then raise again.

Spin-Weighted Formalism

Cosmic Shimmering (arXiv:2410.16274)

- Now apply spin-weighted formalism to distortion matrix in long-distance limit.

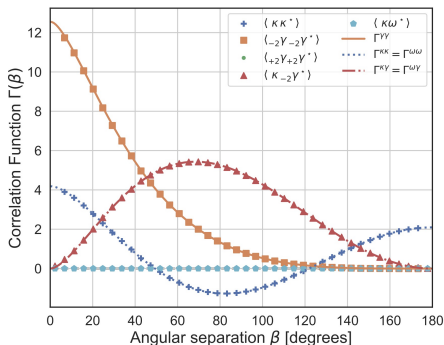


Figure: Cosmic Shimmering Correlators. Credit: Giorgio Mentasti.

Spin-Weighted Formalism

Cross-Correlation (arXiv:2510.25646)

- Novel approach to cross-correlate long-distance and short-distance limits.
- Cross-correlate long-distance PTA redshift with short-distance astrometric observables.

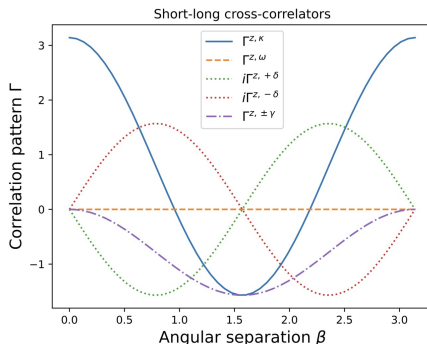


Figure: Cross-Correlators. Credit: Elias Fink.

Signal-to-Noise Ratios

- Need to know how large SNR is for different methods to determine viability.
- Can compare by fixing required SNR and drawing sensitivity curve.

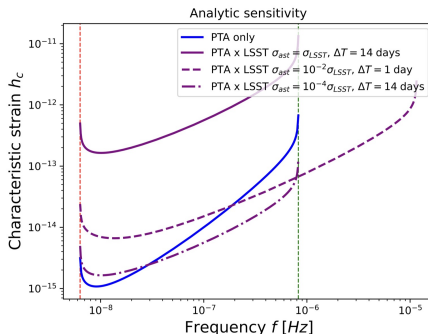


Figure: Sensitivity curves. Credit: Elias Fink.

Future Developments

- Want to extend considerations to non-Einsteinian gravity.
- Can use polarisations and dispersion relations as test of cosmological models through correlation functions.

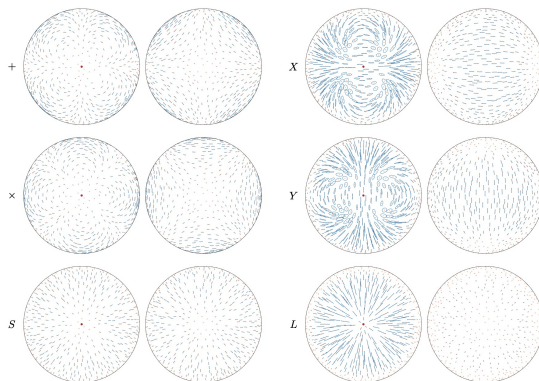


Figure: Non-Einsteinian polarisations. Credit: Mihaylov et. al.